

Solution 1

Question 1 (a)

Probability distribution P over binary variables $x_1, \dots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \dots, x_6) = \frac{1}{Z} \psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)$$

where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

Draw the minimal undirected graph G over which P factors. (This is the undirected graph with the fewest possible edges that contains cliques corresponding to each of the terms in P .)

Solution 1 (a)**Minimal Undirected Graph**

$\therefore$

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Solution 1

Question 1 (b)

Probability distribution P over binary variables $x_1, \dots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \dots, x_6) = \frac{1}{Z} \psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)$$

where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ maximized?

Solution 1 (b)

$\therefore$

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Solution 1

Question 1 (c)

Probability distribution P over binary variables $x_1, \dots, x_6 \in \{0, 1\}$ has the following form:

$$P(x_1, \dots, x_6) = \frac{1}{Z} \psi(x_1, x_2) \psi(x_2, x_3) \psi(x_3, x_4) \psi(x_4, x_5) \psi(x_5, x_6) \psi(x_6, x_1)$$

where for $a, b \in \{0, 1\}$,

$$\psi(a, b) = \begin{cases} 1 & \text{if } a = b \\ 0.5 & \text{otherwise} \end{cases}$$

For what configuration(s) $x \in \{0, 1\}^6$ is $P(x)$ minimized?

Solution 1 (c)

$\therefore$

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Solution 2

Question 2 (a)

In a tournament, each player plays three games. Define three random variables:

- Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
- Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
- Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.

Is C independent of A (i.e., is it true that for any values a, c that A, C might take on, we have $\Pr(A = a, C = c) = \Pr(A = a) \Pr(C = c)$)? Give a brief (one-sentence) reason for your answer.

Hint: The answer is no; show this by exhibiting specific choices of a, c for which $\Pr(A = a, C = c) \neq \Pr(A = a) \Pr(C = c)$.

Solution 2 (a)

$\therefore$

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Solution 2

Question 2 (b)

In a tournament, each player plays three games. Define three random variables:

- Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
- Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
- Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.

Is C conditionally independent of A given B (i.e., is it true that $\Pr(C = c|A = a, B = b) = \Pr(C = c|B = b)$)? Give a brief reason.

Solution 2 (b)

$\therefore$

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Solution 2

Question 2 (c)

In a tournament, each player plays three games. Define three random variables:

- Random variable A is 1 if Tom wins his first game, and is 0 otherwise.
- Random variable B is 1 if Tom wins his first two games, and is 0 otherwise.
- Random variable C is 1 if Tom wins all three of his games, and is 0 otherwise.

Draw an undirected graphical model for (A, B, C) .

Solution 2 (c)

Undirected Graphical Model

∴

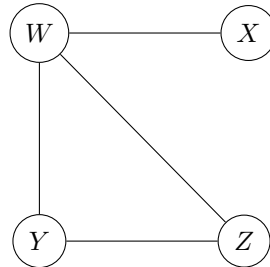
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Solution 3

Question 3 (a)

Four random variables W, X, Y, Z factor over the graph shown below.



What is the functional form of the joint distribution $P(W, X, Y, Z)$?

Solution 3 (a)

∴

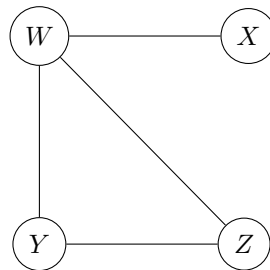
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Solution 3

Question 3 (b)

Four random variables W, X, Y, Z factor over the graph shown below.



Write down a conditional independence property of this distribution.

Solution 3 (b)

$\therefore$

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Solution 4

Question 4 (a)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

One of these variables is a deterministic function of the other two. Identify this relationship.

Solution 4 (a)

$\therefore$

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Solution 4

Question 4 (b)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

Draw the minimal Bayes net G over which this distribution factors.

Hint: Start from (a), which shows that one variable is fully determined by the other two; are the latter two variables independent or not?

Solution 4 (b)

$\therefore$

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Solution 4

Question 4 (c)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

Now show how P factors over G by explicitly giving the three (conditional) probability tables, one per x_i .

Solution 4 (c)

∴

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Solution 4

Question 4 (d)

The joint distribution P over (x_1, x_2, x_3) is given by the following table.

x_1	x_2	x_3	Pr
0	0	0	1/3
0	0	1	1/3
1	0	0	1/6
1	1	1	1/6

Draw the minimal undirected graph G' over which distribution P factors.

Solution 4 (d)

∴

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Solution 5

Question 5 (a)

A Bayesian approach to image denoising. Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).

Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.

Let p denote a specific pixel in the image. Then Y_p is the (reconstructed) value at that pixel, and $Y_{\setminus p}$ is the value at all remaining pixels. Give a precise expression for $\Pr(Y_p = +1 | Y_{\setminus p}, X)$.

Solution 5 (a)

$\therefore$

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Solution 5

Question 5 (b)

A Bayesian approach to image denoising. Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).

Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.

Implement a Gibbs sampler at the appropriate place in the notebook. Initialize Y by setting each pixel to $\{-1, +1\}$ uniformly at random. Then apply the sampler and show the image Y at various stages while the Markov chain is mixing: show any five of the images obtained along the way.

Solution 5 (b)

Python Code: Gibbs sampler

```
1 import numpy as np
2 import pandas as pd
3 from scipy.spatial.distance import cdist
```

Images

∴

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Solution 5

Question 5 (c)

A Bayesian approach to image denoising. Download the file `gibbs.zip` from the course website. This contains a Jupyter notebook and two images (`noisy_bear.png` and `original_bear.png`). The former is a corrupted version of the latter (which is provided only for reference).

Let X be the corrupted image and Y the uncorrupted version that we wish to recover. Both are of the form $\{-1, +1\}^{M \times N}$. In the notebook, a Markov random field joint distribution for (X, Y) is given. You will use this to design a Gibbs sampler for $Y|X$.

To reconstruct the image, just set the value of each pixel p to $\mathbb{E}[Y_p|X]$. Show this image.

Solution 5 (c)

Python Code: Reconstruct Image

```
1 import numpy as np
2 import pandas as pd
3 from scipy.spatial.distance import cdist
```

Reconstructed Image

∴

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Solution 6

Question 6 (a)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Plot the density of f .

Solution 6 (a)

∴

Graduate Level Explanation

Simple Expedited Learning Explanation

Solution 6

Question 6 (b)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Give a suitable proposal distribution $g(x)$. What is a suitable upper bound M on $f(x)/g(x)$?

Solution 6 (b)

∴

Graduate Level Explanation

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Solution 6

Question 6 (c)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Use rejection sampling to generate 10000 samples from f . How many times did you need to sample from g ?

Solution 6 (c)

∴

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Solution 6

Question 6 (d)

Rejection sampling. Suppose we want to sample from a one-dimensional density $f(x)$ for which we do not have a readymade sampler. One way to do this is via rejection sampling. Here's the idea:

- We choose another distribution $g(x)$ from which we know how to sample (e.g. uniform or Gaussian distribution). We will call this the proposal distribution. It is important that $g(x)$ be nonzero whenever $f(x)$ is nonzero.
- Let M be an upper bound on $f(x)/g(x)$; that is, $f(x)/g(x) \leq M$ for all x .
- To generate a sample from f , we repeat the following procedure until it accepts:
 - Generate a sample x from g
 - With probability $f(x)/(Mg(x))$, accept the sample. Otherwise, reject it.

Let's see how this works in practice. Let's say we want to sample a number in the range $[0, \pi]$ from

$$f(x) = \frac{1}{2} \sin x \quad (\text{think of } x \text{ as being in radians})$$

Plot the histogram of the samples you generated, using 50 bins. It should bear a resemblance to the density in (a).

Solution 6 (d)

∴

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